ModelIt!

Teacher's Guide

Why Does Stress Make You Sick?

Modeling Stress Hormones, Immune Suppression, and the Mind-Body Connection

Quick Reference

Grade Level	Grade 12L1
NGSS Standard	HS-LS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS1-3

Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis.

Three-Dimensional Learning

Science Practice	**Planning and Carrying Out Investigations** — Students plan investigations to provide evidence that the HPA axis feedback mechanism maintains cortisol homeostasis during acute stress but fails during chronic activation, leading to immune suppression.
Disciplinary Core Idea	**LS1.A Structure and Function** — The HPA axis is a homeostatic feedback mechanism that maintains cortisol balance: the hypothalamus detects stress, signals the pituitary, which triggers adrenal cortisol release, which feeds back to suppress further activation — a loop that breaks down under chronic stress.
Crosscutting Concept	**Stability and Change** — Students analyze how the stress response system maintains stability (homeostasis) during normal conditions but shifts to a disease-promoting state when chronic stress overwhelms the feedback mechanisms designed for short-term activation.

CAST Testing Alignment

MULTIPLE CHOICE:

A student experiences two weeks of chronic stress during finals with minimal sleep and social isolation. Based on the stress-immune model, which of the following best predicts their health outcome during winter break and why?

CONSTRUCTED RESPONSE:

Using your model, explain the biological mechanism by which strong social support protects immune function during chronic stress. Reference the HPA axis feedback loop, cortisol levels, and at least two specific immune components that are affected.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The field of psychoneuroimmunology — the study of how psychological states affect the nervous and immune systems — has revolutionized our understanding of the mind-body connection. Far from being separate systems, the brain, endocrine system, and immune system are in constant bidirectional communication. Stress hormones directly regulate immune cell production and activity, meaning that your psychological state literally determines how well your body can fight infection and disease.

The HPA Axis: Your Stress Command Center

When the brain perceives a threat, the hypothalamus releases corticotropin-releasing hormone (CRH), which signals the anterior pituitary to release adrenocorticotropic hormone (ACTH) into the bloodstream. ACTH travels to the adrenal glands atop the kidneys, triggering cortisol release. This cascade takes about 20 minutes from threat perception to peak cortisol. Critically, cortisol feeds back to the hypothalamus and pituitary to suppress further CRH and ACTH release — a negative feedback loop that shuts down the stress response once the threat passes. Under chronic stress, this feedback mechanism becomes desensitized, and cortisol remains elevated.

How Cortisol Suppresses Immunity

Cortisol suppresses the immune system through multiple mechanisms: it reduces the production of lymphocytes (T-cells, B-cells) in the thymus and bone marrow, inhibits cytokine signaling between immune cells, reduces the activity of natural killer cells that patrol for cancer and virus-infected cells, and shifts immune response from anti-viral (Th1) to anti-bacterial (Th2), leaving the body more vulnerable to viruses. Acutely, this makes evolutionary sense — during a fight, the body redirects resources from long-term immune surveillance to immediate survival. Chronically, it's catastrophic.

The Social Buffer Effect

Research by Sheldon Cohen, Janice Kiecolt-Glaser, and others has demonstrated that social support physically reduces HPA axis activation. Studies show that people with strong social networks produce less cortisol in response to the same stressor, recover cortisol baseline faster after stress, have higher natural killer cell activity, produce stronger antibody responses to vaccines, and heal from wounds 40% faster. The mechanism appears to involve oxytocin release during positive social interactions, which directly inhibits HPA axis activation.

Allostatic Load and Health Disparities

Bruce McEwen's concept of allostatic load describes the cumulative biological cost of chronic stress. Populations experiencing poverty, discrimination, adverse childhood experiences, and environmental hazards carry disproportionate allostatic load — measurable in higher cortisol, elevated inflammatory markers, accelerated telomere shortening, and premature aging of the immune system. This provides a biological mechanism for observed health disparities: chronic social stress produces chronic immune suppression, which produces higher rates of infection, autoimmune disease, cancer, and cardiovascular disease in specific communities.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify stress exposure, social support, cortisol level, immune function, and health outcome as the key components of the stress-immune system.

E **Establish Relationships**
Students determine that chronic stress elevates cortisol which suppresses immune function, that social support buffers cortisol response, and that the pattern of stress (acute vs. chronic) determines whether immune effects are adaptive or destructive.

V **Visualize & Model**
Students build a computational model showing how the HPA axis feedback loop responds to different stress patterns and how social support modifies cortisol output and downstream immune function.

E **Evaluate Outcomes**
Students run acute stress, chronic stress without support, and chronic stress with support scenarios to test predictions about how stress patterns and social connections affect immune function and health outcomes.

R **Revise & Extend**
Students add sleep quality, inflammatory response, or coping strategy effectiveness to explore how additional factors interact with the basic stress-immune model.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Ever get sick right after finals? That's not coincidence — it's biology. Today we investigate the science behind it."*

Watch For

Show cover image. Ask: How many of you have gotten sick after a stressful period? Quick show of hands.

Slide 2: Learning Objectives

3 min

Say: *"Here's what you'll understand by the end of today — the actual biological mechanism connecting your thoughts to your immune system."*

Watch For

Have students read objectives aloud. Pre-teach vocabulary with endocrine system diagram showing the HPA axis pathway.

Slide 3: Big Question

3 min

Say: *"Your immune system can kill cancer cells. So why can worrying about a test make you catch a cold?"*

Watch For

Quick-write: Students hypothesize a biological mechanism connecting stress and illness. Save to compare with model evidence.

Slide 4: LEVER Framework

2 min

Say: *"We're going to model the hidden conversation between your brain, your hormones, and your immune system."*

Watch For

Preview each LEVER step. Emphasize that this model connects psychology to biology through measurable hormones.

Slide 5: Activity 1: Components

8 min

Say: *"What are the key parts of this stress-immune system? Which inputs come from your environment, and which emerge inside your body?"*

Watch For

Guide sorting of external vs. internal components. Discuss why social support is an external input that modifies internal biology.

Slide 6: Activity 2: Connections

8 min

Say: *"When Stress Exposure is chronic and Social Support is low, what happens to Cortisol Level? And what does sustained cortisol do to Immune Function?"*

Watch For

Students draw arrows showing +/- relationships. Discuss the HPA feedback loop and why chronic stress breaks it.

Slide 7: Activity 3: Simulation

10 min

Say: *"Run each scenario. Notice the dramatic difference between brief stress and sustained stress on your immune system."*

Watch For

Students run all scenarios, record predictions vs. observations. Discuss the social support buffer effect.

Slide 8: Discoveries

5 min

Say: *"Your model reveals that chronic stress literally turns down your immune system — and social connection literally turns it back up."*

Watch For

Class discussion of discoveries. Compare initial hypotheses to model evidence. Emphasize the evolutionary mismatch.

Slide 9: STEM Challenge

12 min

Say: *"Now design a program that targets the specific biological pathways your model revealed. No generic wellness posters."*

Watch For

Groups design evidence-based programs targeting HPA axis regulation and social support. Present with model data justification.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Stress Exposure, Social Support
- **INTERNAL:** Cortisol Level, Immune Function, Health Outcome

Relationships

- **Stress Exposure to Cortisol Level:** POSITIVE (+) — Greater stress exposure activates the HPA axis more intensely and for longer durations, increasing cortisol production and bloodstream concentration.
- **Social Support to Cortisol Level:** NEGATIVE (-) — Stronger social support reduces HPA axis activation through oxytocin-mediated pathways, lowering cortisol output even when stress exposure remains constant.
- **Cortisol Level to Immune Function:** NEGATIVE (-) — Elevated cortisol suppresses lymphocyte production, reduces natural killer cell activity, and impairs antibody response, progressively weakening immune defenses.
- **Immune Function to Health Outcome:** POSITIVE (+) — Higher immune function means greater capacity to fight infections, detect abnormal cells, and heal injuries, directly improving overall health outcomes.

Key Discoveries

- Acute stress actually enhances immune function briefly — cortisol mobilizes immune cells for potential injury, which is why the fight-or-flight response is adaptive for short-term survival
- Chronic stress is immunosuppressive because sustained cortisol exposure reduces lymphocyte production, impairs antibody response, and increases systemic inflammation — the opposite of the acute response
- Social support is not just emotional comfort — it measurably reduces HPA axis activation and cortisol release, creating a biological buffer that partially protects immune function during chronic stress
- There is a lag between stress cessation and immune recovery — this explains why people often get sick right after a stressful period ends, when cortisol drops but immune function hasn't yet restored

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Stress is all in your head — it can't really make you physically sick"	Stress is a full-body physiological event, not just a mental state. The HPA axis converts psychological perception of threat into measurable hormonal changes (cortisol, adrenaline) that directly affect immune cell production, inflammatory pathways, cardiovascular function, and wound healing. Brain scans, blood tests, and immune assays all show physical changes during stress. The mind-body connection is not metaphorical — it operates through specific, measurable neuroendocrine pathways.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Some stress is good for you, so all stress management is unnecessary"	The model clearly distinguishes between acute and chronic stress. Acute stress IS beneficial — it enhances alertness, mobilizes energy, and briefly boosts immune readiness. This is the 'good stress' (eustress) people reference. But chronic stress — where cortisol remains elevated for days, weeks, or months — is unambiguously harmful. The issue isn't eliminating all stress but preventing the chronic activation that damages the HPA feedback loop and suppresses immunity.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Relaxation techniques are too simple to affect real biology"	Evidence-based stress reduction techniques produce measurable biological changes. Mindfulness meditation reduces cortisol by 25% and increases natural killer cell activity. Regular exercise normalizes HPA axis sensitivity and reduces inflammatory markers. Deep breathing activates the parasympathetic nervous system, directly counteracting the sympathetic stress response. These aren't placebo effects — they are physiological modifications of the same neuroendocrine pathways your model shows driving immune suppression.